Alexander Ng

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Experience

Software Engineer III, *Walmart Global Tech* April 2026 – Present

- Engineered core ingestion and feature pipelines for Walmart's large-scale ML-driven inventory placement forecasting platform for all third-party sellers, integrating 18+ upstream data sources supporting nationwide 1-Day delivery decisions
- Built distributed Spark/Scala ETL pipelines processing 180M+ daily forecast records across TB-scale datasets; optimized queries, parallelized transformations, and implemented retry/fallback frameworks to meet strict 6–7 hour production SLAs
- Owned fulfillment enrichment pipelines and end-to-end Airflow orchestration powering automated inventory placement systems, including dependency coordination, cascading failure handling, and backup recovery workflows

Software Engineer II, *Walmart Global Tech* Feb 2024 – April 2026

- Built an automated integration testing framework for data pipelines that generates and executes dependency-aware DAG workflows from declarative configs, reducing development and regression cycles by ~30% (~20 hours per sprint).
- Developed machine learning–driven demand forecasting pipelines on Apache Spark, processing over 90 billion rows of distribution data, leveraging tree-based regression and advanced exponential smoothing to automate predictions and enhance inventory planning.
- Migrated and optimized Google Cloud Storage datasets supporting over 7 million rows of safety stock demand forecasts, enabling faster, more reliable distribution system analytics and replenishment planning built on Airflow + Spark (GCP).
- Led development of an event-driven purchase order management system for distribution operations using Spring Boot microservices on Kubernetes, backed by Kafka, Cassandra, and PostgreSQL; scaled to 1.7M daily orders with optimized batch scheduling to achieve a maximum total daily compute time of under 3 hours.
- Resolved critical Kafka threading/data loss and PostgreSQL indexing infrastructure issues, improving SQL query times by 90% (2 hours → 15 minutes) and preventing 30% of recurring maintenance issues.
- Spearheaded Disaster Recovery compliance across 265 applications, implementing automated failover and backup validation to reduce downtime impact by 8%.

Software Engineer Intern, *Walmart Global Tech* May 2023 – Aug 2023

- Developed a Kafka-based supply chain service with multithreaded consumers for horizontal scaling and high-throughput event ingestion; improved back-pressure handling during peak loads.

Course Staff, *University of California, Berkeley* June 2022 – May 2023

- Conducted lab lectures and office hours on data structures, guiding students through algorithm implementation, complexity analysis, and debugging.

Undergraduate Researcher, *University of California, Berkeley*

Multilingual NLP & Cultural Nuances Sep 2022 – May 2023

- Analyzed transformer-based LLMs (PyTorch) to integrate cultural semantics and context-aware embeddings for early language learning applications. Applied transformer-based architectures and zero-shot evaluations to assess performance across languages, dialects, and cultural registers.

Educational Data Analysis Aug 2021 – Dec 2021

- Analyzed student problem-solving behaviors in parsing-based programming assignments to identify learning patterns and improve instructional strategies.

Projects

Fine-Tuning GPT-2 for Historical Comprehension

- Fine-tuned GPT-2 with Hugging Face and PyTorch to generate historically plausible narratives, improving diversity and reducing overfitting; results published on GitHub: https://github.com/NAYCal/GPT-Histories.

Gitlet

- Java-based version-control system with cryptographic hashing, data persistence, and graph traversals.

NumC

- NumPy-like library in C with optimized array/matrix operations using cache blocking, multiprocessing, and multithreading.

Education

University of California, Berkeley Aug 2021 - Dec 2023

Bachelor of Science in Electrical Engineering and Computer Science

Relevant Coursework: Deep Neural Networks, Machine Learning, Database Systems, Machine Structures, Discrete Maths & Probability, Data Structures, and Data Science.

Skills

Programming Languages: Java, Kotlin, Python, C, C++, RISC-V, SQL

Frameworks/Technologies: Spring Boot, Apache Kafka, Apache Cassandra, Apache Airflow, Apache Spark, MongoDB, PostgreSQL, DB2, GraphQL, REST APIs, JUnit, Flask, PyTorch, Pandas, scikit-learn, Matplotlib, Scala

Tools: Git, GitHub, Jira, Confluence, Kubernetes, Google Cloud Platform, Microsoft Azure